

Reachy Mini Assistant — Built by Claude Code, Runs On-Prem

Big-Brain DEV (Claude Code) builds it efficiently · Small-Brain OPS runs it securely, locally, offline

BIG BRAIN · DEV

Claude Code

agentic software developer

- plans & writes the code
- refactors / fixes bugs
- runs & verifies tests
- commits & pushes to git
- writes docs (this PDF)

Delivered — whole system, iteratively & efficiently:

--local-media backend
(mic / cam / speaker)

deterministic intercepts
(mic / cam / video / status)

plug-in tools
(camera, mic, video, time)

web UI :8090
(text query, toggles, video)

wake-word unmute
(on-device tfllite)

gapless TTS pipeline
(crackle fix)

conversation history
+ text-tool-call recovery

config + this
architecture PDF

Weeks of engineering compressed; every step test-verified and version-controlled.

builds · ships · maintains

SMALL BRAIN · OPS

secure on-prem runtime — local, offline-capable, no cloud

USER

voice / typed
web UI :8090

MEDIA I/O

pluggable:
Reachy | local
(sounddevice/CV)

PERCEPTION

VAD (Silero)
wake-word tfllite
STT faster-whisper

DETERMINISTIC INTERCEPTS

mic / camera / video
status — instant
no LLM · no network

REASONING (only if needed)

local LLM/VLM Ollama
+ tool registry (plug-in)
history · vision

OUTPUT

TTS Kokoro/XTTS (local, pluggable) -> speaker (gapless)
Web UI: tokens · transcript · map · video panel
emotion -> motor (robot)

SECURE / ON-PREM

runs on the device / LAN — no cloud, no telemetry
data never leaves the box · airplane-mode capable
only optional tools (search/APIs) touch the network

EXTENSIBLE

Claude Code adds a
tool / backend / clip
in minutes; config-
driven (settings.yaml)

Same conceptual split, two timescales: the big brain (Claude Code) does the thinking-heavy build; the small brain runs lean, private, and offline in production.